A Detection Method for Tea Leaf Blight in Multialtitude UAV Remote Sensing Images Integrating Style Transfer and Detection Networks

Gensheng Hu, *Member, IEEE*, Zijing Wei, and Yongcheng Jiang

***Abstract*—Tea leaf blight (TLB) is a common disease that is widely found in tea gardens. Accurate detection is the key to preventing TLB. Addressing the issues of dense tea leaves, small TLB spots, and inconsistent target distribution in uncrewed aerial vehicle (UAV) images taken at different altitudes, which lead to low detection accuracy in existing methods, this study proposes a detection method for TLB in multialtitude UAV images by integrating style transfer and detection networks. This method generates fake TLB leaf images using the D-CycleGan style transfer network, which reduces the cost of acquiring TLB datasets and increases the diversity of samples. A multialtitude tiny TLB detection network (MTTDNet) is constructed to accurately detect TLB spots in UAV images of different altitudes. Directional Multi-Conv (DMConv) is introduced into MTTDNet to expand the receptive field and enhance the perception of small-target TLBs. A composite structure block (CSB) is incorporated into the feature fusion stage of MTTDNet to reduce false and missed detections in leaf-dense regions by promoting multidimensional information interaction. In this study, the DNIoU loss function is used to guide the training of MTTDNet by calculating the intersection ratio and center distance of the predicted and target bounding boxes, enabling the network to converge quickly and achieve accurate detection. Compared to the baseline network, MTTDNet improved the AP values for the detection results on datasets taken by a UAV at altitudes of 8, 10, and 14 m by 4.6%, 4.1%, and 6.7%, respectively. Compared with advanced target detection methods such as YOLOv11 and Swin-Transformer, the AP values for the detection results of MTTDNet on the datasets taken at 8, 10, and 14 m were improved by more than 7.7%, 6.1%, and 2.8%, respectively. Hence, the proposed method has important applications in the task of TLB detection in UAV images at different altitudes.**

***Index Terms*—Deep learning, multiscale scene, object detection, style transfer, tea leaf blight (TLB), uncrewed aerial vehicle (UAV).**

Received 12 March 2025; revised 27 May 2025, 20 July 2025, 16 September 2025, and 4 October 2025; accepted 12 October 2025. Date of publication 15 October 2025; date of current version 30 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 32372632, in part by the Major Natural Science Research Projects in Colleges and Universities of Anhui Province under Grant KJ2020ZD03, and in part by Hefei Municipal Natural Science Foundation under Grant HZR2411. *(Corresponding authors: Gensheng Hu; Yongcheng Jiang.)*

Gensheng Hu and Zijing Wei are with the National Engineering Research Center for Agro-Ecological Big Data Analysis and Application, Anhui University, Hefei, Anhui 230601, China (e-mail: hugs2906@sina.com; 2843884601@qq.com).

Yongcheng Jiang is with the School of Electrical Engineering and Automation, Anhui University, Hefei, Anhui 230601, China (e-mail: 05047@ahu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3621626

I. Introduction

TEA leaf blight (TLB) is a common disease that occurs in tea gardens around the world, especially in humid, warm environments. Tea leaves infected with TLB gradually lose moisture, become dry and curled, and eventually fall off. In severe cases, the disease may spread to the branches of the tea plant, causing the branches to wither or even the whole tea plant to die, seriously affecting the yield and quality of tea. Pesticide spraying is one of the effective methods to control TLB. However, overspraying of pesticides will damage the ecological environment. Therefore, efficient detection of TLB in tea gardens is important to ensure the high quality of the tea and high tea yields and can avoid pesticide misuse [1].

Traditional crop disease detection methods rely on manual field investigations, which are costly, inefficient, and have poor timeliness. In recent years, deep learning and computer vision techniques have achieved extraordinary success in crop disease detection tasks. Guan [2] developed a new plant disease detection method by stacking four CNN networks (Inception, ResNet, Inception-ResNet, and DenseNet) and achieved a detection accuracy of 87% on a publicly available dataset of 10 plant species and 61 healthy and diseased plant leaves. Khan et al. [3] proposed a two-stage apple disease detection system based on deep learning. First, diseases were classified by a lightweight classification network, and single classes of spots were then detected and localized separately based on the classification results, resulting in a detection accuracy of 88% and a mean average precision (mAP) value of 42%. Karthik et al. [4] proposed an attention-based deep residual network for detecting tomato leaf diseases that achieved an accuracy of 98% on a tomato leaf dataset. Dananjayan et al. [5] developed a new citrus diseased leaf dataset, CCL'20, and evaluated the detection performance of different CNN architectures on it in terms of computational cost and inference speed. Shin et al. [6] tested the performance of detection networks such as AlexNet, SqueezeNet, GoogLeNet, ResNet-50, SqueezeNet-MOD1, and SqueezeNet-MOD2 on a powdery mildew (PM) disease dataset and comprehensively evaluated the detection accuracy and computational cost of different networks. Xu et al. [7] proposed a real-time target detection network called Pruned-YOLO v5s+Shuffle (PYSS) for melon leaf disease; their approach improved the mAP value and precision of the detection results by 6.2% and 6.4%, respectively, and reduced the network size and inference time by

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

85% and 7.5%, respectively, compared to YOLOv5. Arun and Umamaheswari [8] proposed a complete concatenated deep learning (CCDL) architecture, which can detect and classify crop diseases without learning them. They achieved a higher detection accuracy of 98.14% using a smaller network with a size of 10 MB. These methods improve the efficiency of crop disease detection, save a lot of labor costs, and provide a paradigm for crop disease detection tasks.

Most of the above methods have utilized high-resolution images taken on the ground to detect different types of crop diseases. Under actual production conditions, these detection methods, however, offer limited coverage and suffer from low data acquisition efficiency, consequently preventing their practical application in large-scale agricultural disease control. It is currently common in academia to use remote sensing images as dataset samples, thus enabling the large-scale detection of crop diseases. Due to its advantages of low cost and easy operation, UAV remote sensing technology has been widely used for crop disease detection. Wu et al. [9] used a UAV to collect a large number of canopy images of early infected diseased pine trees to construct a training dataset and trained and tested the Faster-RCNN and YOLO networks for detection on this dataset. Ishengoma et al. [10] designed a hybrid convolutional neural network combining VGG16 and InceptionV3 for the detection of maize leaf disease in UAV images and found that this hybrid network reduced the training time by 16%–44% compared to other networks while achieving a high detection accuracy. Lei et al. [11] used a UAV to capture multispectral data to estimate the severity of betel nut yellow leaf disease by combining five models: backpropagation neural network (BPNN), decision tree, naïve Bayes, support vector machine (SVM), and k-nearest neighbors classification. Deng et al. [12] proposed a novel detection framework called RustQNet for quantitative assessments of the disease index of wheat stripe rust. Their framework fused multimodal UAV images through the designed mutual information minimization (MIM) module to obtain higher detection accuracy. Rodriguez et al. [13] detected potato late blight using multispectral images acquired by a UAV. The UAV images were first preprocessed to remove soil and weeds, and the dataset was then tested using random forest, gradient boosting classifier, support vector classifier, linear support vector classifier, and k-nearest neighbors classifier. In this way, the effectiveness of different models in terms of detecting and evaluating potato late blight was verified. Zeng et al. [14] proposed a rubber tree PM detection method based on UAV low-altitude remote sensing and hyperspectral imaging technology. They extracted the spectral reflection wavelength littoral from the region of interest of the UAV hyperspectral image to evaluate the detection effect of different spatial resolution images. Compared with the detection of crop diseases using images captured on the ground, the method of using UAV images further expands the coverage area of detection and improves the detection efficiency.

Although the application of UAV imagery effectively improves the detection area of crop pests and diseases and enhances the timeliness of detection, the high flight altitude of a UAV means that capturing images in this way will inevitably cause the loss of some detailed information. Researchers have proposed various methods to solve the problem of low detection accuracy due to the loss of detail in UAV images. Kisantal et al. [15] used oversampling to reduce the induction of large targets into the detection network and copy-pasted small targets multiple times into each image to fill in the absence of positional information on the small targets in the image, thus improving the network's detection accuracy for small targets. Rabbi et al. [16] designed an edge-enhanced super-resolution network called EESRGAN to improve the quality of remotely sensed images. These high-quality images were used for testing the detection network to improve the detection accuracy of small targets. Deng et al. [17] constructed an extended feature pyramid network (EFPN), which included super-resolution pyramid layers for the extraction of image details and acquisition of high-resolution features to enhance the network's ability to capture small targets. Yang et al. [18] proposed a network called QueryDet in which shallower high-resolution features were introduced to boost small-target detection while ensuring computational lightness. Chen et al. [19] proposed a TinyDet network that used average pooling to mitigate the alignment bias caused by downsampling in the backbone network and applied FPN to fuse multilayer feature information. TinyDet outperformed the SOTA network in terms of detection accuracy for both small and medium targets on the COCO dataset. Through the processing of UAV images, the accuracy of detection is further improved.

Detection of TLB from UAV images is more challenging than the detection of other crop diseases. The background of the tea garden is very complex, and factors such as soil, dead leaves, and overlapping leaves can significantly interfere with feature extraction from the target; in addition, the small size of TLB spots means that they can be easily overlooked, resulting in a decrease in detection accuracy. In response to these problems, researchers have proposed a number of methods. Hu et al. [20] and Jiang et al. [21] combined super-resolution and detection networks to obtain high-resolution features for diseased spots using a super-resolution network and detected these spots with a modified detection network, resulting in more accurate detection results. Bao et al. [22] proposed a DDMA-YOLO detection network to coordinate channel attention (CA) and spatial attention by fusing DDMA modules in the backbone to reduce false and missed detections due to dense leaf distribution. Jiang et al. [23] designed an MLDNet detection network to compensate for the information loss due to illumination by fusing RGB images with simulated infrared images. MLDNet also contained an SR branch that reconstructed high-resolution features from low-resolution features extracted in the backbone, thus guiding the detection network to learn small-target features more effectively. These methods provide a reference for TLB detection tasks. However, there are still many challenges in the TLB detection task for UAV images.

Due to the different flight altitudes of UAVs, the feature distributions of TLBs and the backgrounds of images taken at different altitudes may show large differences; in addition, collecting large numbers of remote sensing images of TLBs at different altitudes and labeling them involves large

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

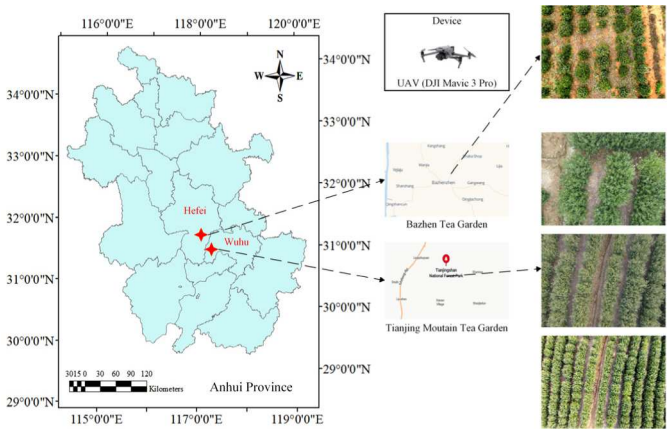


Fig. 1. Data acquisition site.

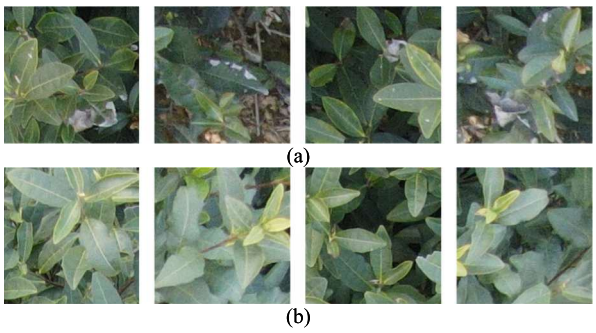


Fig. 2. (a) TLB and (b) healthy leaf sample images after cropping.

human and material costs. In this study, the style transfer technique is integrated into the task of detecting TLBs in UAV images, and a method for detecting TLBs from UAV images taken at different altitudes is proposed that combines style transfer and detection networks. This method reduces the cost of TLB dataset acquisition and increases the number and diversity of sample images by transferring TLB spot features to healthy leaf images through a style transfer network. A model called multialtitude tiny TLB detection network (MTTDNet) is designed to overcome the problems of background complexity and small-target detection in UAV images. MTTDNet fully extracts the multiscale information in UAV images and achieves accurate detection of small-target TLBs in UAV images at different altitudes.

The main contributions of this study are as follows.

1) Style transfer network D-CycleGan is used to increase the number and diversity of the sample images and reduce the cost of TLB dataset acquisition.

2) Detection network MTTDNet is designed to detect TLBs from UAV images at different altitudes and reduce the false and missed detections.

3) The DNIoU loss function is adopted to guide MTTDNet in learning TLB spot features to achieve accurate detection.

4) The proposed method integrating style transfer and detection networks can effectively detect TLBs in UAV images taken at different altitudes.

II. Materials and Methods

A. Data Acquisition

The data used in this study were collected from Tianjing Mountain Tea Garden (located in the southwestern part of Wuhu, Anhui, 31°14'N, 117°36'E, 40 m above sea level) and Bazhen Tea Garden (located in Hefei, Anhui, 31°16′ N, 117°36′ E, 40 m above sea level). Fig. 1 shows the data acquisition site. The data collection device was a DJI Mavic 3 Pro UAV equipped with a 48-Mpixel medium and telephoto camera. The data on Tianjing Mountain Tea Garden were acquired in the afternoon of December 24, 2024, in clear weather, and the UAV flew at altitudes of 4, 8, 10, and 14 m. The data on Bazhen Tea Garden were collected in the afternoon of May 3, 2021, in clear weather, and the UAV flew at an altitude of 10 m. The size of each captured image was 8000 × 6000. The acquired images were divided based on the UAV flight altitude, with 65 original images taken at an altitude of 4 m, 20 taken at 8 m, 15 taken at 10 m, and 10 taken at 14 m. In addition, UAV images at different altitudes were collected in Tianjing Mountain Tea Garden for robustness analysis of the network on cloudy days. These images were captured on December 6, 2023, November 4, 2023, and February 11, 2025, with 15 images taken at an altitude of 8 m, 14 images taken at an altitude of 10 m, and 13 images taken at an altitude of 14 m. The size of each captured original image was 8000 × 6000.

B. Data Preprocessing

1) Data Preprocessing for D-CycleGan: In this study, the style transfer technique was used to transfer the TLB spot features into the healthy leaf images in order to obtain fake TLB images at different altitudes and in different scene environments. The D-CycleGan style transfer network will be described in detail in Section II-D. Training of D-CycleGan required real TLB leaf images and healthy leaf images as samples. In order to learn high-quality TLB spot features, TLB leaf images and healthy leaf images taken at an altitude of 4 m were used as training samples for D-CycleGan: a total of 300 TLB leaf images and 300 healthy leaf images, all of size 416 × 416, were cropped from original UAV images of size 8000 × 6000 taken at an altitude of 4 m to construct the training sample set. Some examples of the cropped TLB leaf sample images and healthy leaf sample images are shown in Fig. 2. The original UAV images containing healthy leaves taken at altitudes of 8, 10, and 14 m were cropped to a size of 416 × 416 and fed into the trained D-CycleGan to obtain fake TLB leaf images, which were used as the training and validation sets for the detection network.

2) Data Preprocessing for MTTDNet: The detection network MTTDNet will be described in detail in Section II-E. The training and validation sets for the MTTDNet consist of labeled fake TLB leaf images generated by D-CycleGAN. In order to avoid labeling errors caused by overlapping leaves, only the TLB spots were labeled in this study rather than the entire diseased leaves. In addition, in order to avoid the problem of overlapping detection boxes due to the density of small-sized disease spots, a single detection box was used for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

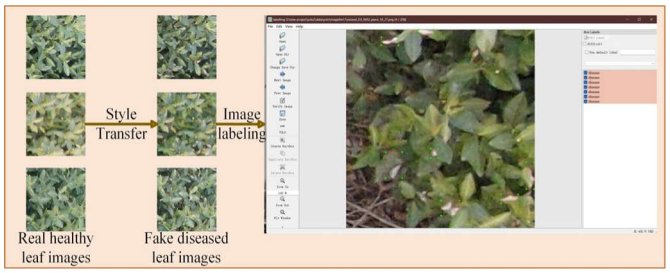


Fig. 3. Data preprocessing for the detection network.

TABLE I
NUMBERS OF TRAINING SAMPLES AFTER DATA PREPROCESSING

Altitude	Training set	Validation set
8m	300	100
10m	300	100
14m	300	100

labeling the dense areas of disease spots. The process of data preprocessing for the detection network is shown in Fig. 3. A total of 1200 fake TLB leaf images at altitudes of 8, 10, and 14 m were obtained through style transfer and were divided into the training and validation sets for the MTTDNet network in a ratio of 3:1. The numbers of training and validation samples for MTTDNet after data preprocessing are shown in Table I.

C. Steps in the TLB Detection Process

This study presents a two-stage detection method for TLBs from UAV images based on style transfer and detection networks. The first stage involved the training of the D-CycleGan network and the generation of fake TLB images at different altitudes. Images of TLB and healthy leaves taken at an altitude of 4 m were used as samples to train D-CycleGan to extract high-quality TLB spot features. Healthy leaf images taken at altitudes of 8, 10, and 14 m were input to the trained D-CycleGan to generate fake TLB leaf images, which were used to construct training and validation samples for the detection network MTTDNet. The second stage involved training and testing of the MTTDNet network. The network was trained and validated using fake TLB leaf images at different altitudes generated by D-CycleGan, and then, real TLB images taken at different altitudes were fed to the trained MTTDNet network to realize the accurate detection of TLB disease.

The specific steps in this process were as follows.

Step 1 (Data Preprocessing for D-CycleGan): Original images taken at an altitude of 4 m were cropped to obtain 300 images of TLB leaves and 300 images of healthy leaves, all with a size of 416 × 416.

Step 2 (Training of D-CycleGan): The healthy leaf images were fed into the generator to get the fake TLB leaf images, and the generator parameters were adjusted by calculating the loss function through the discriminator. Similarly, the TLB leaf images were fed to the generator to obtain fake images of healthy leaves, and the generator parameters were adjusted by calculating the loss function through the discriminator. Using the cycle consistency loss function, the generator was trained to generate TLB leaf images.

Step 3 (Data Preprocessing for MTTDNet): A total of 1200 healthy leaf images of size 416 × 416 were cropped from UAV images taken at 8, 10, and 14 m. These images were fed into the trained D-CycleGan to output the fake TLB leaf images, which were then labeled and divided in a ratio of 3:1 to obtain the training and validation sets for the detection network MTTDNet.

Step 4 (Training of MTTDNet): Training and validation sets were input into MTTDNet to train and validate the network. At the feature extraction stage, the receptive field was expanded by Directional Multi-Conv (DMConv) to reduce the small-target TLB loss during downsampling. At the feature fusion stage, image edges were enhanced and multidimensional information fusion was facilitated by introducing a composite structure block (CSB).

Step 5 (Testing of MTTDNet): The original UAV images taken at 8, 10, and 14 m were cropped to a size of 416 × 416 and fed to the trained MTTDNet to accomplish TLB detection from real TLB leaf images taken at different altitudes.

The complete TLB detection process for UAV images taken at different altitudes is shown in Fig. 4, where the green arrow indicates the training and testing process of D-CycleGan and the black arrow indicates the training and testing process of MTTDNet.

D. D-CycleGan Network

A style transfer network can effectively extract the features of disease spots and render these spots on healthy leaf images collected at different altitudes and in different scenes, to give fake TLB leaf images that are similar to real TLB images. The structure of the D-CycleGan style transfer network used in this study is shown in Fig. 5. A healthy leaf image is input to the generator G_D to produce a fake TLB leaf sample, and the identity of this sample is determined by the discriminator D_D. Similarly, a TLB leaf image is input to the generator G_H to generate a fake healthy leaf sample and the identity of this sample is determined by the discriminator D_H. The generator parameters are adjusted by the losses fed back from the discriminator to allow the generator to produce more images.

The healthy and TLB leaf images were divided into two different domains, X and Y. Bidirectional mapping was realized by two generators and two discriminators, and a cyclic consistency loss was introduced to ensure the validity and reversibility of the transformation. The generator G_D was used for transformation from domain X to domain Y ($G_D : X \rightarrow Y$), while the generator G_H was used for transformation from domain Y to domain X ($G_H : Y \rightarrow X$). The discriminators D_D and D_H were used to evaluate the realism of the generated samples. To avoid a situation where generators produce completely unrelated outputs, D-CycleGan applies a cyclic consistency loss, meaning that if an image from domain X is fed into generator G_D to be transferred into domain Y and then transferred back to domain X through generator G_H, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

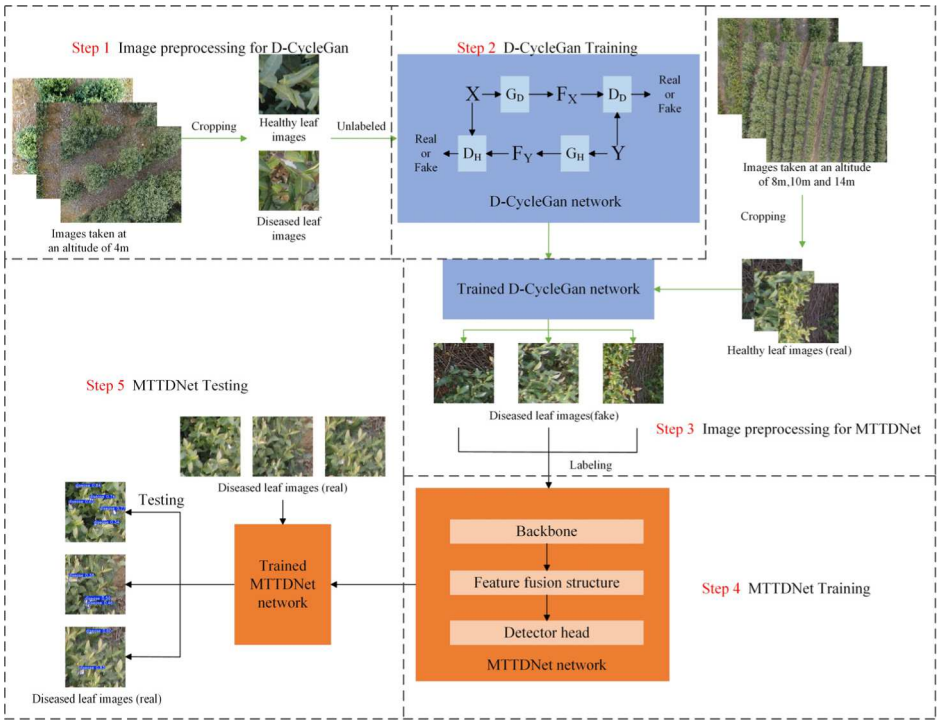


Fig. 4. Complete process of the TLB detection method.

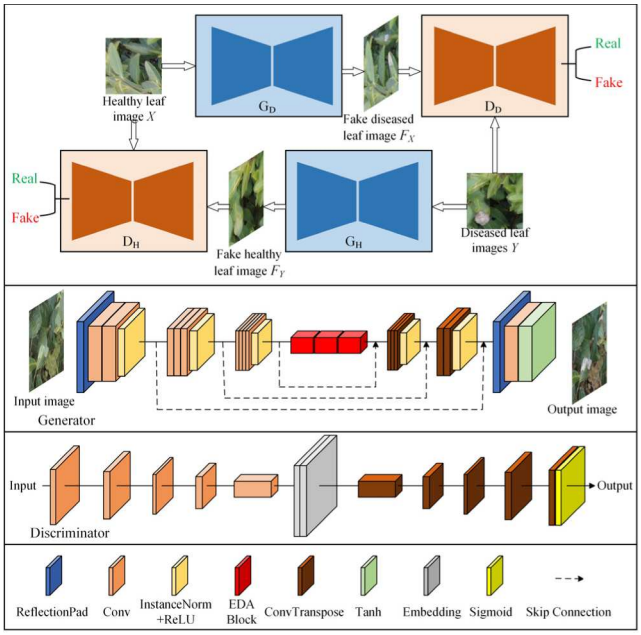


Fig. 5. Structure of the D-CycleGan network.

final result should be as close as possible to the original input image. The process can be expressed as follows:

$$x \approx G_H\left(G_D\left(x\right)\right) \tag{1}$$

$$y \approx G_D\left(G_H\left(y\right)\right). \tag{2}$$

The total loss L_{total} of D-CycleGan consists of two parts: the adversary loss L_{GAN} and the cyclic consistency loss L_{cyc}. In the case of the generator G_D and discriminator D_D, the formulas

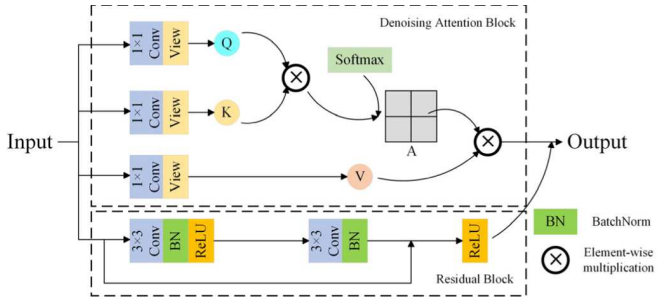


Fig. 6. EDA block.

for the adversary loss L_{GAN} and the cyclic consistency loss L_{cyc} can be expressed as follows:

$$L_{\text{GAN}}\left(G_D, D_D, X, Y\right) = E_{y\sim p_{\text{data}}(y)}\left[\log D_D\left(y\right)\right] + E_{x\sim p_{\text{data}}(x)}\left[\log\left(1 - D_D\left(G_D\left(x\right)\right)\right)\right] \tag{3}$$

$$L_{\text{cyc}}\left(G_D, G_H\right) = E_x\left[\left\|G_H\left(G_D\left(x\right)\right) - x\right\|_1\right] + E_y\left[\left\|G_D\left(G_H\left(y\right)\right) - y\right\|_1\right] \tag{4}$$

where E denotes the expected value; $p_{\text{data}}(x)$ and $p_{\text{data}}(y)$ represent the real data distribution for domains X and Y, respectively; and $\|*\|_1$ denotes the L_1 norm. Hence, the total loss L_{total} of D-CycleGan can be expressed as

$$L_{\text{total}} = L_{\text{GAN}}\left(G_D, D_D, X, Y\right) + L_{\text{GAN}}\left(G_H, D_H, Y, X\right) + L_{\text{cyc}}\left(G_D, G_H\right). \tag{5}$$

Equation (5) balances the adversary loss and cyclic consistency loss to make the images generated by the network similar in appearance and consistent in detail, thereby generating more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Algorithm for D-CycleGan Training

Input: Healthy leaf image X, Diseased leaf image Y, numbers of iterations K, batch size m.
Output: Trained generator G_D.

1: **for** k = 1 to K **do**
2: **Train** G_D;
3: Draw F_X from G_D, discriminate whether the image is true or false.
4. Update the G_D via gradient-based optimizer based on Eq.(5).
5. **Train** G_H;
6. Draw F_Y from G_H, discriminate whether the image is true or false.
7. Update the G_H via gradient-based optimizer based on Eq.(5).
8: **end for**

ideal output results. Algorithm 1 presents the pseudocode used to train D-CycleGan.

Due to the high flight altitude of the captured UAV images, there are some areas of noise and blurring in the images. InstanceNorm and ReLU are therefore introduced after the convolutional layer of D-CycleGan's generator to stabilize the style of the input image and to preserve the original details of the image. In addition, to enhance the denoising effect, three parallel EDA blocks are introduced to the generator. As shown in Fig. 6, the EDA block consists of a denoising attention block (DAB) and a residual block (RB). DAB enhances the representation of feature maps through the self-attention mechanism. It overlays the results of RB to focus the image on important feature regions and improve the denoising effect, which is calculated as follows:

$$A = \text{Softmax}\left(\frac{Q \cdot K}{\sqrt{C'}}\right) \tag{6}$$

$$X = \text{ReLU}\big(\text{BN}\big(\text{Conv}_{3\times3}\big(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}\big(\text{Input}\big)\right)\right)\right)\big) + \text{Input}\big) \tag{7}$$

$$\text{Output} = A \cdot V + X \tag{8}$$

where C' denotes the scaling factor, each element of A represents the strength of the association between different locations on the feature map, and X denotes the output feature-enhanced result of the input feature map after RB. By fusing the residual connection structure and the self-attention mechanism, the noise in the image is effectively removed.

Fig. 7 gives some examples of the resulting images after style transfer. It can be seen that D-CycleGan successfully renders realistic TLB spots on healthy leaf images, thus providing high-quality samples for training of the detection network.

E. MTTDNet Detection Network

1) Overview of MTTDNet: In this study, a detection network MTTDNet is designed to detect TLBs from UAV images of different altitudes. The overall structure of MTTDNet is shown in Fig. 8. MTTDNet can be mainly divided into three parts: feature extraction, feature fusion, and the detector head.

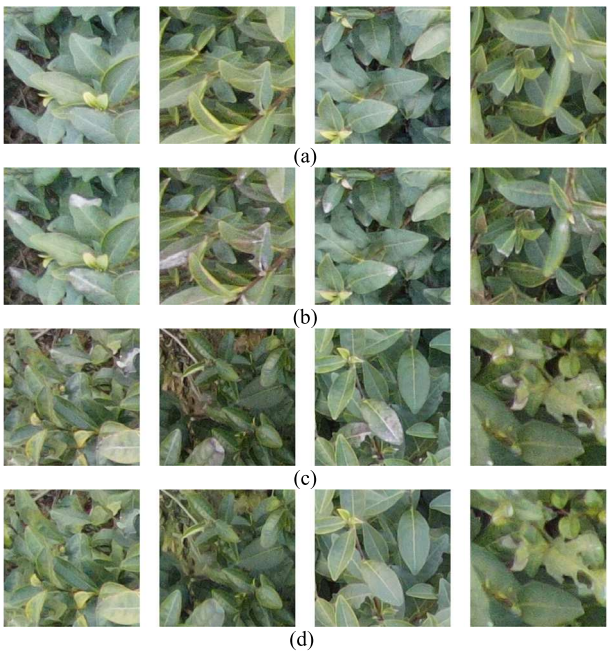


Fig. 7. Examples of real images of healthy and TLB leaves and generated fake images. (a) Real healthy leaf images. (b) Fake TLB leaf images. (c) Real TLB leaf images. (d) Fake healthy leaf images.

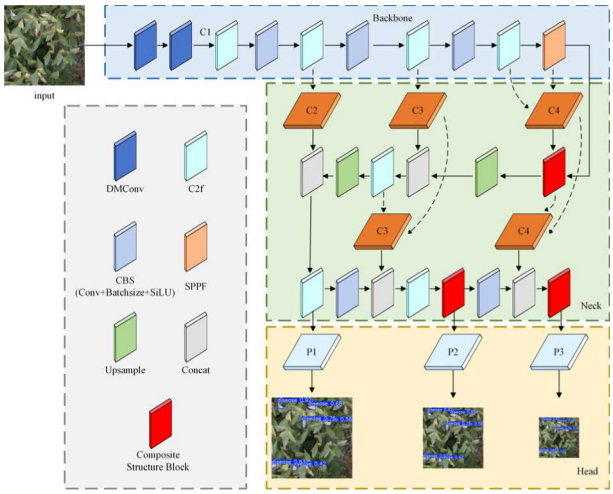


Fig. 8. Structure of MTTDNet.

In the feature extraction part, two DMConv layers were introduced into the backbone of MTTDNet to expand the receptive field, aggregate feature information from multiple directions, and reduce the loss of small-target TLB features due to downsampling.

In the feature fusion part, three CSB modules are introduced. The CSB module consists of the EEA and EMDA modules, which enhance the edge features and retain more detailed information about the small-target TLBs. In addition, the EMDA module promotes the fusion of CA and pixel attention (PA), which reduces false and missed detections of TLB spots due to dense areas of leaves.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Algorithm for MTTDNet Training

Input: Fake TLB leaf images $X = (x_1, y_1),..., (x_N, y_N)$, which generated by D-CycleGan, numbers of iterations K, batch size m.
Output: Trained MTTDNet.
1: Split X into training set X_t and validation set X_v with a ratio of 3:1.
2: Initialize parameters of MTTDNet θ_0.
3: **for** k = 1 to K **do**
4: Train MTTDNet with X_t.
5: **for** $x_{vi} \in x_v$ **do**
6: Predict the labels of x_{vi}, y_{pi}.
7: Compare the predicted values with ground truth values using Eq.(27).
8: Update θ_0 to θ_k.
9: **end for**
10: **end for**

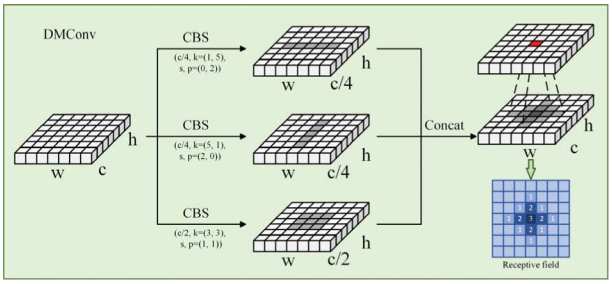


Fig. 9. Structure of DMConv.

In the detector head, the DNIoU loss function is used to train MTTDNet. This loss function improves the accuracy of localization by calculating the Wasserstein distance between the predicted and the target boxes so that the network pays more attention to the exact matching of the position and size of the bounding box during the learning process. In addition, by calculating the intersection and IoU of the predicted and the target boxes and the Euclidean distance between their centers, the predicted box is guided to move quickly toward the target box to speed up convergence.

Algorithm 2 presents the pseudocode used to train MTTDNet.

2) DMConv: In order to better extract the small-target TLB features from UAV images and reduce the information loss in the downsampling process, DMConv is introduced into the backbone of MTTDNet to enhance the network's ability to extract feature information in different directions. The structure of DMConv is shown in Fig. 9. The input tensor $X \in R^{h\times w\times c}$ extracts the feature information in different directions via a 1×5 convolution in the horizontal and vertical directions and a 3×3 convolution to obtain the tensors $Y_1 \in R^{h\times w\times c/4}$, $Y_2 \in R^{h\times w\times c/4}$, and $Y_3 \in R^{h\times w\times c/2}$. The output tensors are channel concatenated to obtain the output tensor Y, and the normalization and activation functions are introduced to obtain the output result Y_{output}. The specific formulas are as follows:

$$Y_1 \in R^{h\times w\times \frac{c}{4}} = \text{Conv}_{1\times 5}(X) \tag{9}$$

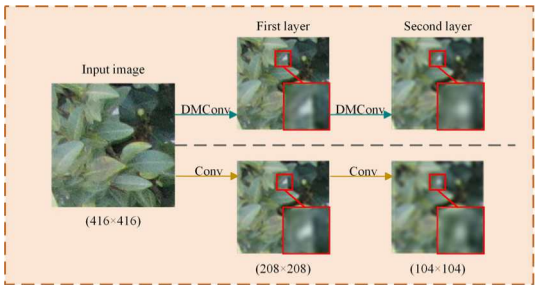


Fig. 10. Visualization of DMConv and Conv output results.

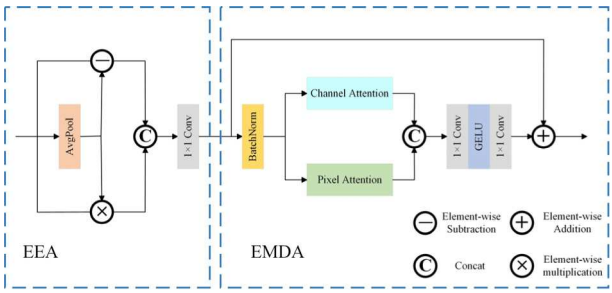


Fig. 11. Structure of CSB.

$$Y_2 \in R^{h\times w\times \frac{c}{4}} = \text{Conv}_{5\times 1}(X) \tag{10}$$

$$Y_3 \in R^{h\times w\times \frac{c}{2}} = \text{Conv}_{3\times 3}(X) \tag{11}$$

$$Y \in R^{h\times w\times c} = \text{Cat}(Y_1, Y_2, Y_3) \tag{12}$$

$$Y_{\text{output}} \in R^{h\times w\times c} = \text{SiLU}(\text{BN}(Y)) \tag{13}$$

where h, w, and c denote the height, width, and number of channels for the image, respectively; Cat denotes channel concatenation; BN denotes batch normalization; and SiLU represents the activation function. The risk of network overfitting is reduced by introducing normalization and activation functions after the concatenated feature maps, and nonlinearities are introduced into the network to mitigate the problems of gradient explosion and gradient vanishing [24].

In order to visualize the feature extraction effect of DMConv, the results after downsampling using DMConv and Conv are shown in Fig. 10. It can be seen that by expanding the receptive field and capturing features in different directions, DMConv enhances the network's ability to extract edge structures, thus retaining more feature information about small-target TLBs and providing more high-quality learnable features for small-target TLB detection.

3) CSB: Due to the small sizes of the TLB spots in UAV images, they are easily overlooked, resulting in a large number of missed detections. In addition, in areas of dense tea leaves, the network may not separate the edges of the disease spots well due to mutual occlusion of the leaves, resulting in false detection. Hence, the extraction of edge information from the feature map and the enhancement of the interaction of feature information in different dimensions need to be considered at the feature fusion stage. Based on the above analysis, a CSB module is introduced at the feature fusion stage of MTTDNet.

As shown in Fig. 11, the CSB module consists of an EEA module and an EMDA module, which are used to enhance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

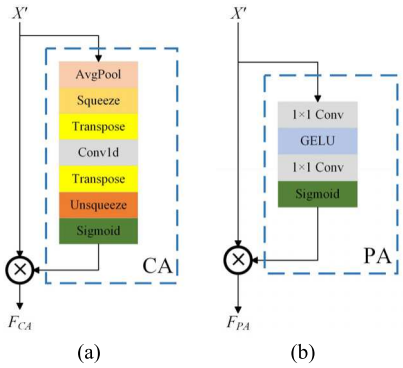


Fig. 12. Schematics showing (a) CA and (b) PA.

the edge features of the disease spots and to aggregate the multidimensional information, respectively.

In the EEA module, the input feature map X is subjected to a 3×3 average pooling layer and then undergoes element-by-element subtraction and element-by-element multiplication with the original map to separate the high-frequency feature map X_h and the low-frequency feature map X_l, respectively. The high- and low-frequency feature maps are concatenated, and a 1×1 convolution is then used to facilitate the interaction of channel information to obtain the output feature map Y'. The specific formulas are as follows:

$$X_1 = \text{AvgPool}_{3\times3}(X) \tag{14}$$

$$X_h = X - X_1 \tag{15}$$

$$X_l = X \odot X_1 \tag{16}$$

$$Y' = \text{Conv}_{1\times1}(\text{Cat}(X_1, X_2)) \tag{17}$$

where $-$ denotes the element-by-element subtraction and $\odot$ denotes the element-by-element multiplication. This simple structure effectively enhances the edges of the feature map for the detection of small-target TLB.

CA and PA modules are fused in the EMDA module to enhance the interaction of information on multidimensional features through a parallel approach [25], thus reducing the false and missed detections in leaf-dense areas.

The structures of CA and PA are shown in Fig. 12. The feature map Y' is input to the EMDA module, processed using batch normalization, and then input to the two branching modules CA and PA to extract the channel feature map F_{CA} and the pixel-level feature F_{PA}, which are concatenated and then fused through an MLP structure. It is finally added element by element to the input feature map Y' to get the output result Y. The specific formulas are as follows:

$$X' = \text{BN}(Y') \tag{18}$$

$$F_{\text{CA}} = \text{Sig}\left(\text{Unsqueeze}\left(\text{Tr}\left(\text{Conv1d}\left(\text{Tr}\left(\text{Squeeze}\right)\right)\right)\right)\right) \times \left(\text{Avg}_{3\times3}(X')\right) \tag{19}$$

$$F_{\text{PA}} = \text{Sig}\left(\text{Conv}_{1\times1}\left(\text{GELU}\left(\text{Conv}_{1\times1}(X')\right)\right)\right) \tag{20}$$

$$Y = Y' + \text{MLP}(\text{Cat}(F_{\text{CA}}, F_{\text{PA}})) \tag{21}$$

where Sig denotes the sigmoid function, Tr denotes the transposition, Conv1d denotes the 1-D convolution, and $+$ denotes the element-by-element summation. CA effectively captures the global contextual information of the entire feature map through the global pooling operation and dynamically generates weights based on the importance of each channel to make the network more focused on important features. PA allows the network to make fine-grained adjustments to each pixel location by amplifying important localized areas in the image, thus better capturing subtle differences in the image. By combining the two kinds of attention, EMDA enables the network not only to identify important features at the macro level but also to precisely localize and enhance these features at the micro level, which greatly enhances the expressiveness and flexibility of the network.

4) DNIoU: Due to the small sizes of TLB spots and their susceptibility to interference from dense leaves and other complex backgrounds, an appropriate loss function is crucial to guide MTTDNet to learn higher quality image features. NWDLoss [26] calculates the similarity of Gaussian distribution obtained from the Wasserstein distance measure for the predicted and the target boxes. In the case where there are no areas of overlap, a similarity assessment can still be performed by calculating the Wasserstein distance. In addition, for multiscale targets, the Wasserstein distance maintains its robustness to the size of the detected target and avoids missed detections due to the small TLB spots. For predicted box $A = (cx_a, cy_a, w_a, h_a)$ and target box $B = (cx_b, cy_b, w_b, h_b)$, the NWDLoss is calculated as follows:

$$W_2^2(N_a, N_b) = \left\| \left(\left[cx_a, cy_a, \frac{w_a}{2}, \frac{h_a}{2} \right]^{\text{T}}, \left[cx_b, cy_b, \frac{w_b}{2}, \frac{h_b}{2} \right]^{\text{T}} \right) \right\|_2^2 \tag{22}$$

$$\text{NWD}(N_a, N_b) = \exp\left(-\frac{\sqrt{W_2^2(N_a, N_b)}}{C} \right) \tag{23}$$

$$L_{\text{NWD}} = 1 - \text{NWD}(N_a, N_b) \tag{24}$$

where $W_2^2(N_a, N_b)$ denotes the distance metric between the predicted box A and the target box B. This cannot be used directly as a similarity metric and is therefore exponentiated to obtain a new metric, the normalized Wasserstein distance (NWD). L_{NWD} is the final expression of NWDLoss. In addition, DIoU takes into account the region of overlap between the target box and predicted box as well as the distance between the centroids of the two boxes, thus guiding the network to converge to the position of the target box during the training process. Even if there is no area of overlap between the target box and predicted box, DIoU can guide the direction of convergence of the predicted box by calculating the distance between the centroids of the two boxes [27]. The specific formulas for DIoU are as follows:

$$\text{DIoU} = \text{IoU} - \frac{\rho^2(b, b^t)}{c^2} \tag{25}$$

$$\text{IoU} = \frac{|\text{B} \cap B^t|}{|B \cup B^t|} \tag{26}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

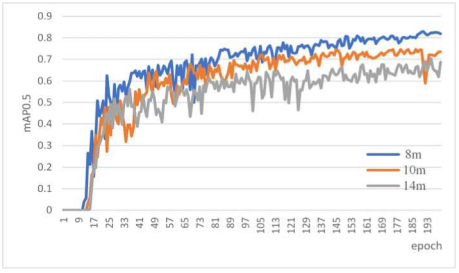


Fig. 13. Training curves for different datasets.

where B and B^t denote the predicted box and target box, respectively; b and b^t denote the center coordinates of the predicted box and target box, respectively; IoU denotes the intersection ratio between the target and predicted boxes; ρ stands for the Euclidean distance between b and b^t; and c represents the diagonal distance of the smallest closure region that can contain both the target and predicted boxes.

In this study, the above two loss functions are combined to form a DNIoU loss function that guides the training of the MTTDNet network for better detection of TLBs from UAV images taken at different altitudes. The DNIoU loss function is calculated as

$$\text{DNIoU} = w * L_{\text{NWD}} + (1 - w) * \text{DIoU} \tag{27}$$

where w is the weight ratio of the two loss functions NWD-Loss and DIoU. DIoU mainly calculates the intersection and IoU of the predicted and the target boxes and guides the prediction box to move quickly toward the target box, whereas NWDLoss mainly measures the difference in distance between the predicted and the target boxes. By combining the two loss functions, the information on the two metrics (overlap ratio and distance) is fully utilized to better guide the feature learning of MTTDNet. The results show that the best performance is achieved when the DNIoU loss function is used for the task considered in this study.

Fig. 13 shows the mAP@0.5 curves for the training results of the MTTDNet network on the 8-, 10-, and 14-m datasets. It can be seen that the network stabilizes after 100 epochs, indicating that the network has largely converged.

III. Experimental Results and Analysis

A. Experimental Configuration and Evaluation Metrics

Experiments were conducted using the Ubuntu 20.04LTS operating system, with the PyTorch 1.5 deep learning framework. The other components used were an Intel 8375C CPU and dual NVIDIA GeForce RTX 3080Ti GPUs, with CUDA 10.1 and cuDNN 7.6.5 as GPU gas pedals, and the scripting language for the experiment was Python 3.8. The parameter settings for the style transfer network and the detection network in training are shown in Table II.

In order to assess the quality of the generated fake TLB leaf images, two evaluation metrics are introduced in this study: FID [28] and IS [29]. The specific formulas for their calculation are as follows:

$$\text{FID} = \left|\mu_r - \mu_g\right|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{1/2}\right) \tag{28}$$

TABLE II
Experimental Parameter Settings

Experimental parameter	Style transfer network	Detection network
Input size	416×416×3	416×416×3
Learning rate	0.0003	0.0001
Epoch	200	200
Batch size	2	16
Momentum	0.5	0.937
Weight_decay	0.0002	0.0005
Optimization algorithm	SGD	Adam

$$\text{IS} = \exp\left(E_x D_{\text{KL}}\left(p\left(y\,|\,x\right)\|p\left(y\right)\right)\right) \tag{29}$$

where μ_r and Σ_r represent the mean and covariance matrices of the feature vectors extracted from the real image, respectively; μ_g and Σ_g represent the mean and covariance matrices of the feature vectors extracted from the generated image, respectively; Tr denotes the trace of the matrix, E_x denotes the expected value about all generated images x; and $D_{\text{KL}}(p(y\,|\,x)\|p(y))$ is the Kullback–Leibler scatter between the conditional probability distribution $p(y\,|\,x)$ and the marginal probability distribution $p(y)$, which is used to measure the difference between the two distributions. FID measures the quality of the generated images by calculating the similarity of the distribution of the generated images with the real images in the feature space. The lower the FID, the closer the distribution of the generated images is to the distribution of the real images, which indicates that the quality of the generated image is higher and more realistic. IS evaluates the performance of the generative network by measuring the diversity and clarity of the generated images. The higher the IS, the better the visualization, and the higher the diversity.

In order to evaluate the detection effectiveness of the proposed method, the precision, recall, and mAP are used as evaluation metrics. These metrics are calculated as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{30}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{31}$$

$$\text{mAP} = \frac{\sum_{i=1}^{n} \text{AP}_i}{n} \tag{32}$$

where true positive (TP) is the number of samples correctly predicted as TLB (correct detections) and false positive (FP) is the number of samples in which the background was incorrectly predicted as TLB spots (false detections). False negative (FN) is the number of TLB spots incorrectly predicted as background (missed detections), and true negative (TN) is the number of background samples correctly predicted (correct detections). Precision is the probability that a sample predicted as positive is indeed positive, while recall is the probability that a sample that is actually positive is correctly predicted to be positive. mAP provides a comprehensive way of evaluating the network performance by focusing not only on the performance of a single metric but also on evaluating

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.



Fig. 14. Examples of fake TLB leaf images generated by different style transfer networks. (a) Original image. (b) StyleGAN. (c) CycleGan. (d) CAP-VSTNet. (e) Puff-net. (f) DEADiff. (g) D-CycleGan.

TABLE III
DETECTION RESULTS FROM TRAINING SAMPLES GENERATED BY DIFFERENT STYLE TRANSFER NETWORKS AND IMAGE QUALITY ASSESSMENT

Style transfer network	8m/10m/14m			FID	IS
	Precision(%)	Recall(%)	mAP(%)		
StyleGAN	64.2/58.7/52.3	68.1/64.3/59.4	67.2/61.1/56.2	27.5	6.8
CycleGan	82.6/79.4/**69**	76.3/68.1/66.6	84.5/76.4/66.9	12.9	9.5
CAP-VSTNet	73.5/68.2/62.7	71.9/67.3/65.5	73.7/70.4/63.8	25.8	5.8
Puff-net	75.3/70.2/60.3	72.1/62.4/60.9	72.7/64.4/59.3	18.4	7.3
DEADiff	72.6/69.5/62.2	71.9/64.7/61.1	71/66.7/61.4	19.5	6.6
D-CycleGan	**86.8**/**82**/68.2	**80.2**/**70.5**/**68**	**87.5**/**79.4**/**69.4**	**11.3**	**10.4**

the average performance of the network across all metrics. The mAP value is high only if precision and recall are both high.

B. Experiments on Different Style Transfer Networks

In order to validate the effectiveness of the D-CycleGan network designed in this study in terms of generating TLB samples for the TLB detection task, experiments were conducted to compare the detection results for the images generated using D-CycleGan with the results for images generated by other style transfer networks. The experimental results are shown in Table III and Fig. 14. It can be seen that StyleGAN [30] pays insufficient attention to the TLB spots and prefers to render the color information in the original image; this means that the generated image contains fewer spots and does not provide the detection network with enough information about the features of the TLBs, leading to low detection accuracy. CycleGan [31] and CAP-VSTNet [32] can effectively distinguish the difference between healthy and TLB leaves by focusing on the features of the TLB spots and rendering them onto the healthy leaf images. However, since the samples used in this study were taken by UAVs flying at high altitudes, there are some areas of blurring and noise in the images; these factors lead to some shadows in the images generated by CycleGan and CAP-VSTNet and are not conducive to the detection network learning the feature information of the TLBs and the background, meaning that they affect the detection accuracy. The images generated by Puff-net [33] and DEADiff [34] have poor ability to recover details. Especially, DEADiff has learned a lot of color features of TLB, but has poor ability to learn the shape of disease spots, resulting in a large distribution of generated disease spots, which is greatly different from the actual situation. D-CycleGan introduces an EDA block in the generator to denoise the generated image, and skip connections are implemented in the generator structure to preserve the edge features of the original image. The FID and IS scores are superior to the other networks, which means that the quality of the generated images is higher. Therefore, D-CycleGan can provide high-quality feature information for the learning of the detection network.

In order to illustrate the specific effect of the training samples generated by D-CycleGan, Fig. 15 shows the detection results obtained by training the detection network on the images generated by different style transfer networks and then using the trained network for testing. In Fig. 15, the red boxes represent detected TLB spots, the blue boxes represent missed detections, and the yellow boxes represent false detections or areas where the position of the predicted box deviates dramatically from the position of the target box. It can be seen that the images generated by D-CycleGan provide high-quality TLB features, enabling the detection network to more accurately detect TLBs from UAV images taken at different altitudes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

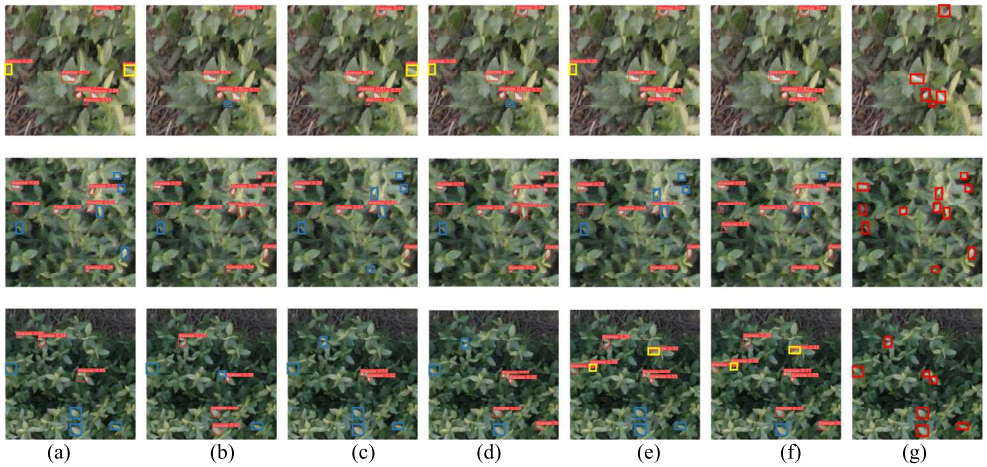


Fig. 15. Detection results using fake TLB leaf images generated by different style transfer networks as training samples. (a) Original image. (b) StyleGAN. (c) CycleGan. (d) CAP-VSTNet. (e) Puff-net. (f) DEADiff. (g) D-CycleGan.

TABLE IV
COMPARISON OF DETECTION RESULTS FROM DIFFERENT DETECTION NETWORKS

Detection network	8m/10m/14m			Size(MB)
	Precision(%)	Recall(%)	mAP(%)	
YOLOv5	76.5/74.8/59.2	72.3/69/58.9	78.2/74.9/63.5	**5.2**
YOLOv6	76.7/75.3/66.5	69.1/65.4/56.5	79.1/74/64.3	8.5
YOLOv8	81.7/75.4/59.6	67.6/69.4/58.4	82.9/75.3/62.7	6.1
YOLOv9	80.3/70.1/66	77.5/67.3/65.9	79.8/73.3/66.5	106.5
YOLOv10	71.1/64.8/59.3	68.1/59.7/57.7	74.7/65.6/59.3	8.0
YOLOv11	78.7/74.2/**69**	66.9/70.1/60.6	77.2/75.6/66.6	5.4
Faster R-CNN	65.8/60.3/55.1	63.7/58.2/54.3	64.8/57.9/54	124.3
Swin-Transformer	71.8/69.2/63.5	72.3/67.8/63.2	73.5/69/64.4	275.1
SuperYOLO	78.6/73.5/60.4	73.5/68.4/57.8	74.4/70.2/58.3	13.5
UAV-DETR	74.3/69.6/58.1	70/66.3/56.4	72.1/68.3/57.4	35.4
MTTDNet	**86.8**/**82**/68.2	**80.2**/**70.5**/**68**	**87.5**/**79.4**/**69.4**	7.3

C. Experiments on Different Detection Networks

In order to evaluate the performance of the proposed MTTDNet detection network, this section compares the detection results of MTTDNet with those of other state-of-the-art detection networks and describes ablation experiments carried out for each component of MTTDNet.

1) Comparison Experiments With Other Detection Networks: To demonstrate the effectiveness and superiority of MTTDNet on the task of TLB detection from UAV images taken at different altitudes, we compare the detection results from nine classical and state-of-the-art detection networks under the same configuration: YOLOv5, YOLOv6 [35], YOLOv8, YOLOv10 [36], YOLOv11 [37], Faster R-CNN [38], Swin-Transformer [39], SuperYOLO [40], and UAV-DETR [41]. The detection results are shown in Table IV and Fig. 16. From the detection results, it can be seen that the YOLO network has insufficient feature extraction capability for small-target TLBs, resulting in a large number of missed detections of small-target TLBs. In addition, the YOLO network's inability to extract edge features leads to a large number of overlapping anchor frames in some areas of leaf overlap. Faster R-CNN and Swin-Transformer are also unable to effectively extract the feature information of small-target TLBs in UAV images, and the generated weight files take up a large amount of memory, which is not conducive to deployment in actual detection devices. The small-target detection methods, SuperYOLO and UAV-DETR, are poorly adapted to scale variations in UAV images taken at different altitudes, resulting in low detection accuracy for datasets taken at 14-m altitude. The MTTDNet network proposed in this study expands the receptive field of the network by the designed DMConv and gradually stretches with the increase of distance, which learns more distinguishable features for the network and strengthens the feature extraction capability for small-target TLBs. In addition, the introduced CSB module enhances the multidimensional feature interaction of the image by fusing the channel information and pixel information, which reduces the phenomenon of overlapping anchor boxes caused by overlapping areas. This allows for a balance between detection accuracy and computational cost.

2) Ablation Experiments: To verify the contributions of the proposed DMConv, CSB, and DNIoU on the experimental results, ablation experiments were designed. The experimental results are shown in Table V. It can be seen that by introducing DMConv, the receptive field of the network is enlarged and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

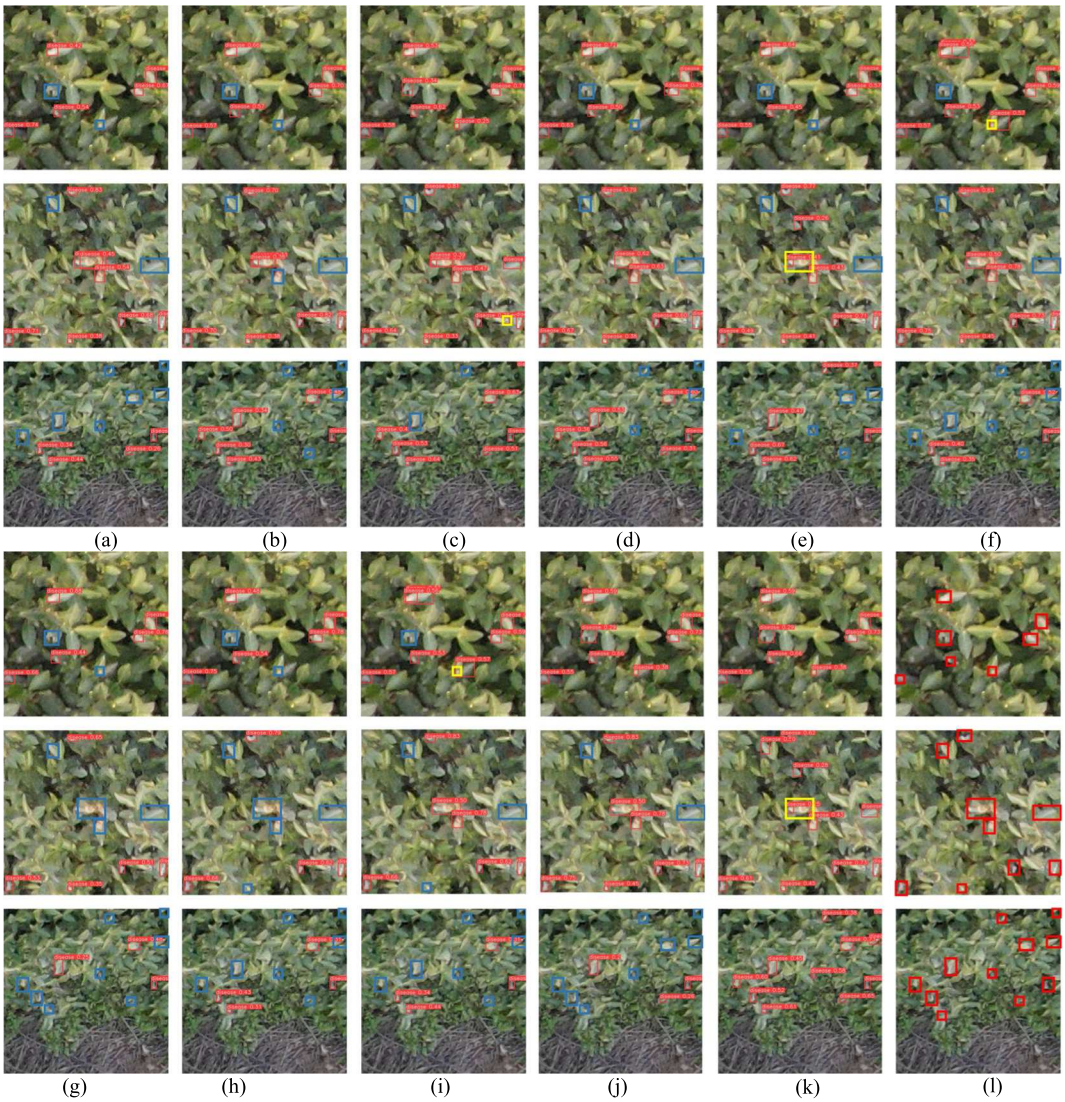


Fig. 16. Examples of detection results from different networks. (a) YOLOv5. (b) YOLOv6. (c) YOLOv8. (d) YOLOv9. (e) YOLOv10. (f) YOLOv11. (g) Faster R-CNN. (h) Swin-Transformer. (i) SuperYOLO. (j) UAV-DETR. (k) MTTDNet. (l) GT.

TABLE V
DETECTION RESULTS FROM THE ABLATION EXPERIMENTS

DMConv	CSB	DNIoU	8m/10m/14m		
			Precision(%)	Recall(%)	mAP(%)
-	-	-	81.7/75.4/59.6	67.6/69.4/58.4	82.9/75.3/62.7
√	-	-	75.6/75.5/63.7	77.8/71.7/60.1	82.9/77.3/65.1
-	√	-	81.6/76/61.4	72.9/69.8/61.3	81.7/74.5/66.6
-	-	√	80.8/76.2/62.1	73.5/71.4/63.8	81.6/76.3/64
√	√	-	83.5/78.1/66.4	75/**72.2**/64.2	84.1/76.5/65.7
√	-	√	84.4/79.1/67	77.3/69.2/65.5	83.8/75.4/67.3
-	√	√	82.6/78.5/65.9	76.5/66.7/61.3	83.7/77.9/65.8

the feature extraction ability of the backbone of MTTDNet is enhanced, which effectively improves its ability to capture small-target TLBs.

As the shooting altitude increases, the improvement in detection accuracy compared to the original network is more pronounced, with the mAP value increasing by 2% and 2.4% on the 10- and 14-m datasets, respectively. The introduction of CSB effectively reduces the overlapping of anchor frames caused by the dense areas of leaves and reduces the false detection rate, with the *recall* improved by 5.3%, 0.4%, and 2.9% on the 8-, 10-, and 14-m datasets, respectively. DNIoU guides the prediction box to move quickly toward the target

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

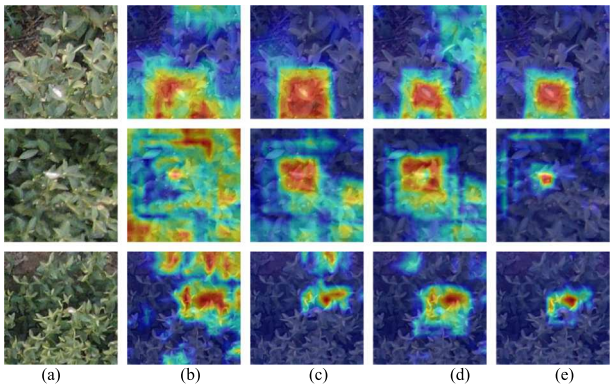


Fig. 17. GradCAM visualization results for ablation experiments with DMConv and CSB. (a) Original images. (b) No introduced. (c) With DMConv. (d) With CSB. (e) DMConv and CSB.

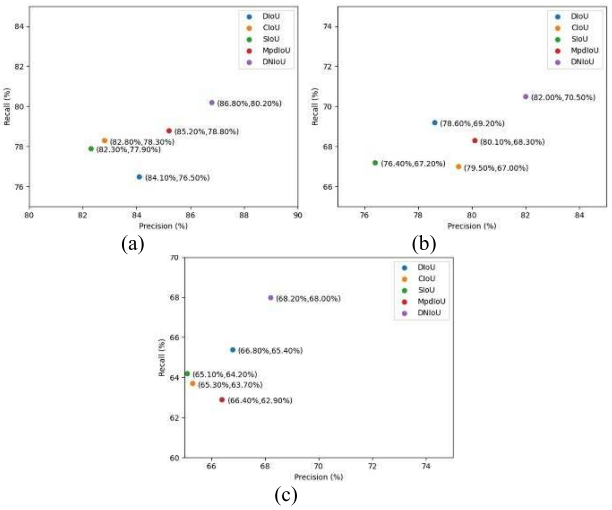


Fig. 18. Detection results using different loss functions. (a) 8, (b) 10, and (c) 14 m.

box by calculating the intersection ratio and center point distance during the training process, which effectively improves the detection accuracy and accelerates the convergence speed.

In order to illustrate the role of the proposed DMConv and CSB in a more intuitive way, some GradCAM visualizations of the results generated in the ablation experiments are shown in Fig. 17. It can be seen that by introducing DMConv and CSB, the level of attention paid by MTTDNet to the detected small-target TLBs is significantly improved, which helps to guide MTTDNet to learn high-quality TLB features in order to improve the detection accuracy of small-target TLBs.

3) Comparative Experiments on Loss Functions: In order to improve the detection accuracy of small-target TLBs and to increase the convergence speed, the DNIoU loss function is used in this study, which guides the predicted box to move quickly toward the target box by calculating the intersection ratios and centroid distances. To demonstrate the effectiveness and superiority of the DNIoU, an experiment was conducted to compare the results obtained using several other loss functions to guide network training: DIoU [27], CIoU [42], SIoU [43], and MpdIoU [44]. The results are shown in Fig. 18. It can be seen that the DNIoU loss function achieves the highest

TABLE VI
NUMBERS OF TEST SAMPLES FOR ROBUSTNESS ANALYSIS EXPERIMENTS

Datasets	Altitude		
	8m	10m	14m
December 6, 2023	125	151	74
November 4, 2023	187	124	124
February 11, 2025	65	110	96

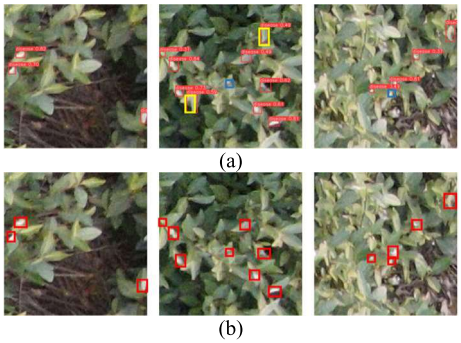


Fig. 19. Some detection results of robustness analysis experiments. (a) Detection results. (b) Ground truth.

precision and recall on the UAV dataset containing images taken at different altitudes, which suggests that the DNIoU can efficiently guide the training of MTTDNet, performing well on the task of detecting small-target TLBs.

4) Robustness Experiment: Images in the dataset used for the robustness analysis of the network were collected in Tianjing Mountain Tea Garden on December 6, 2023, November 4, 2023, and February 11, 2025 under cloudy weather conditions. These test images containing TLBs were also cropped, and the size of each cropped image is 416 × 416. The number of test samples at different altitudes is shown in Table VI.

The detection results of robustness experiment are shown in Table VII and Fig. 19. It can be seen that the method proposed in this study can still achieve effective detection results on different datasets. As the shooting was conducted under cloudy weather conditions, this confirms that the proposed method has good light adaptability.

IV. DISCUSSION

This study proposes a TLB detection method that combines style transfer with detection networks. The D-CycleGan style transfer network is integrated with the MTTDNet detection network to realize the detection of TLBs in UAV images obtained at different altitudes. The experimental results show that the proposed method achieves the best detection results on datasets taken at altitudes of 8, 10, and 14 m.

Since UAV images containing TLBs must be collected from tea gardens that are infected by the disease, this imposes restrictions on the timing of data acquisition. The problem of the high cost of acquiring TLB images was overcome by designing a D-CycleGan style transfer network that learns TLB spot features from diseased leaf images and renders them onto healthy leaf images. In order to prove that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
DETECTION RESULTS OF ROBUSTNESS ANALYSIS EXPERIMENTS

Datasets	8m/10m/14m		
	Precision(%)	Recall(%)	mAP(%)
December 6, 2023	82.5/78.4/67.2	81.6/69.3/69.1	88.2/80.5/69.8
November 4, 2023	79.8/76.9/65.4	80.3/79.2/68.7	81.3/78.4/66.2
February 11, 2025	85.4/82.2/69.1	81/77.1/68.2	84.4/79.8/69.2

TABLE VIII
DETECTION RESULTS FOR DIFFERENT DATASETS

Datasets	8m/10m/14m		
	Precision(%)	Recall(%)	mAP(%)
Real TLB leaf images	87.3/81.5/68.7	81.6/69.3/69.1	88.2/80.5/69.8
Generated fake TLB leaf images	86.8/82/68.2	80.2/70.5/68	87.5/79.4/69.4

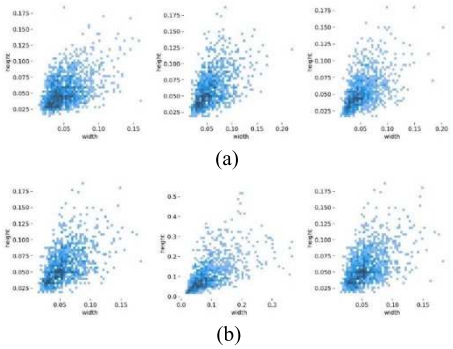


Fig. 20. Distribution of TLB size in different datasets obtained at altitudes of 8, 10, and 14 m, from left to right. (a) Distribution of TLB size in generated fake TLB leaf images. (b) Distribution of TLB size in real TLB leaf images.



Fig. 21. Examples of detection results using different methods. (a) Hu et al. [20]. (b) Jiang et al. [21]. (c) Proposed method.

generated images can achieve the effect of approximating real images, 400 real TLB leaf images taken by UAV at altitudes of 8, 10, and 14 m were selected for this study; the training and validation sets were divided in the ratio 3:1 and then used to train MTTDNet. The size distribution of the TLBs in the generated and real TLB leaf image datasets is shown in Fig. 20. It can be seen that the size distribution for the generated TLB leaf images is similar to that in the real images. From the detection results in Table VIII, it can be seen that the detection accuracy of the network obtained when the generated fake TLB leaf images were used as training samples is only slightly lower than for the network obtained using real TLB leaf images as training samples. This demonstrated that the fake TLB leaf images generated using the D-CycleGan style transfer network represent good-quality training samples. This way of acquiring a dataset greatly reduces the cost of data acquisition and enables training of the TLB detection network in the laboratory using healthy leaf images, thereby overcoming the limitations on sample collection time.

To address the issue of overlap caused by the small size of TLB spots and dense leaves in UAV images, a detection network called MTTDNet was proposed in this study. In this network, DMConv was introduced into the backbone to expand the receptive field and reduce the loss of small-target TLB feature information during feature extraction. In addition, CSB was introduced at the feature fusion stage to enhance the edges via the EEA module, and then, CA and PA were fused by the EMDA module to minimize missed and false detections in leaf-dense areas. Finally, the network convergence and the ability to sense small-target TLBs were accelerated by designing a DNIoU loss function.

In recent years, scholars have proposed a variety of methods for detecting TLBs in UAV images. Hu et al. [20] and Jiang et al. [21] used a super-resolution approach combined with a detection network, which gave excellent performance on UAV datasets. In this study, the detection results of the proposed method are compared with the state-of-the-art TLB detection methods on UAV datasets taken at different altitudes. From the results in Table IX and Fig. 21, it can be seen that the proposed method achieves the best detection results, with good scale adaptation for UAV images taken at different altitudes.

However, the methodology presented in this research remains untested under challenging environmental conditions, including intense illumination, rain, and fog. This limitation stems from the style transfer network's current inability to generate sufficiently high-quality fake TLB leaf images. Consequently, improving the robustness of detection algorithms across diverse weather conditions emerges as a critical research priority for subsequent investigations. Future work should focus on developing adaptive detection networks capable of maintaining detection accuracy despite atmospheric interference and illumination fluctuations of real-world agricultural environments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
DETECTION RESULTS FOR DIFFERENT METHODS

Methods	8m/10m/14m		
	Precision(%)	Recall(%)	mAP(%)
Hu et al. [20]	78.5/74.3/65.2	73.9/68.7/63.1	75.4/72.2/63.7
Jiang et al. [21]	76.4/72.8/66.5	72.6/69.2/65.3	76.5/71.8/64.2
Proposed method	**86.8/82/68.2**	**80.2/70.5/68**	**87.5/79.4/69.4**

V. CONCLUSION

TLB is a common disease that occurs in tea gardens. Accurate detection is the key to preventing TLB and ensuring tea yield and quality. This study has presented a method for detecting TLBs in UAV images taken at different altitudes by integrating style transfer and detection networks. The style transfer method provides a paradigm to cope with the high cost of TLB dataset acquisition. The MTTDNet detection network is designed for the problems of scale inconsistency, small-target detection and leaf overlap in the TLB detection task in UAV images at different altitudes. The proposed method facilitates the interaction of multidimensional feature information by introducing CSB at the feature fusion stage. The experimental results show that the proposed method achieves mAP values of 87.5%, 79.4%, and 69.4% for detection results on datasets taken at 8-, 10-, and 14-m altitudes, respectively, showing significant performance improvement compared to other methods. The proposed method is therefore an effective strategy for the detection of TLBs from UAV images taken at different altitudes. The current work has not yet overcome the impact of extreme weather conditions such as rain and fog, as well as the flight altitude of UAV on detection tasks, which are issues that need to be further addressed in the future.

REFERENCES

[1] G. Hu, H. Wu, Y. Zhang, and M. Wan, "A low shot learning method for tea leaf's disease identification," *Comput. Electron. Agricult.*, vol. 163, Aug. 2019, Art. no. 104852, doi: 10.1016/j.compag.2019.104852.

[2] X. Guan, "A novel method of plant leaf disease detection based on deep learning and convolutional neural network," in *Proc. 6th Int. Conf. Intell. Comput. Signal Process. (ICSP)*, Apr. 2021, pp. 816–819, doi: 10.1109/ICSP51882.2021.9408806.

[3] A. I. Khan, S. M. K. Quadri, S. Banday, and J. L. Shah, "Deep diagnosis: A real-time apple leaf disease detection system based on deep learning," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107093, doi: 10.1016/j.compag.2022.107093.

[4] R. Karthik, M. Hariharan, S. Anand, P. Mathikshara, A. Johnson, and R. Menaka, "Attention embedded residual CNN for disease detection in tomato leaves," *Appl. Soft Comput.*, vol. 86, Jan. 2020, Art. no. 105933, doi: 10.1016/j.asoc.2019.105933.

[5] S. Dananjayan, Y. Tang, J. Zhuang, C. Hou, and S. Luo, "Assessment of state-of-the-art deep learning based citrus disease detection techniques using annotated optical leaf images," *Comput. Electron. Agricult.*, vol. 193, Feb. 2022, Art. no. 106658, doi: 10.1016/j.compag.2021.106658.

[6] J. Shin, Y. K. Chang, B. Heung, T. Nguyen-Quang, G. W. Price, and A. Al-Mallahi, "A deep learning approach for RGB image-based powdery mildew disease detection on strawberry leaves," *Comput. Electron. Agricult.*, vol. 183, Apr. 2021, Art. no. 106042, doi: 10.1016/j.compag.2021.106042.

[7] Y. Xu et al., "Real-time object detection method of melon leaf diseases under complex background in greenhouse," *J. Real-Time Image Process.*, vol. 19, no. 5, pp. 985–995, Oct. 2022, doi: 10.1007/s11554-022-01239-7.

[8] R. A. Arun and S. Umamaheswari, "Effective multi-crop disease detection using pruned complete concatenated deep learning model," *Expert Syst. Appl.*, vol. 213, Mar. 2023, Art. no. 118905.

[9] B. Wu et al., "Application of conventional UAV-based high-throughput object detection to the early diagnosis of pine wilt disease by deep learning," *Forest Ecology Manage.*, vol. 486, Apr. 2021, Art. no. 118986, doi: 10.1016/j.foreco.2021.118986.

[10] F. S. Ishengoma, I. A. Rai, and S. R. Ngoga, "Hybrid convolution neural network model for a quicker detection of infested maize plants with fall armyworms using UAV-based images," *Ecological Informat.*, vol. 67, Mar. 2022, Art. no. 101502, doi: 10.1016/j.ecoinf.2021.101502.

[11] S. Lei, J. Luo, X. Tao, and Z. Qiu, "Remote sensing detecting of yellow leaf disease of arecanut based on UAV multisource sensors," *Remote Sens.*, vol. 13, no. 22, p. 4562, Nov. 2021, doi: 10.3390/rs13224562.

[12] J. Deng et al., "RustQNet: Multimodal deep learning for quantitative inversion of wheat stripe rust disease index," *Comput. Electron. Agricult.*, vol. 225, Oct. 2024, Art. no. 109245, doi: 10.1016/j.compag.2024.109245.

[13] J. Rodríguez, I. Lizarazo, F. Prieto, and V. Angulo-Morales, "Assessment of potato late blight from UAV-based multispectral imagery," *Comput. Electron. Agricult.*, vol. 184, May 2021, Art. no. 106061, doi: 10.1016/j.compag.2021.106061.

[14] T. Zeng et al., "Early detection of rubber tree powdery mildew using UAV-based hyperspectral imagery and deep learning," *Comput. Electron. Agricult.*, vol. 220, May 2024, Art. no. 108909.

[15] M. Kisantal, Z. Wojna, J. Murawski, J. Naruniec, and K. Cho, "Augmentation for small object detection," 2019, *arXiv:1902.07296*.

[16] J. Rabbi, N. Ray, M. Schubert, S. Chowdhury, and D. Chao, "Small-object detection in remote sensing images with end-to-end edge-enhanced GAN and object detector network," *Remote Sens.*, vol. 12, no. 9, p. 1432, May 2020, doi: 10.3390/rs12091432.

[17] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022, doi: 10.1109/TMM.2021.3074273.

[18] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13658–13667, doi: 10.1109/CVPR52688.2022.01330.

[19] S. Chen et al., "TinyDet: Accurate small object detection in lightweight generic detectors," 2023, *arXiv:2304.03428*.

[20] G. Hu, R. Ye, M. Wan, W. Bao, Y. Zhang, and W. Zeng, "Detection of tea leaf blight in low-resolution UAV remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, no. 5601218, pp. 1–18, 2024, doi: 10.1109/TGRS.2023.3339765.

[21] Y. Jiang, Z. Wei, and G. Hu, "Detection of tea leaf blight in UAV remote sensing images by integrating super-resolution and detection networks," *Environ. Monitor. Assessment*, vol. 196, no. 11, p. 1044, Nov. 2024, doi: 10.1007/s10661-024-13221-w.

[22] W. Bao, Z. Zhu, G. Hu, X. Zhou, D. Zhang, and X. Yang, "UAV remote sensing detection of tea leaf blight based on DDMA-YOLO," *Comput. Electron. Agricult.*, vol. 205, Feb. 2023, Art. no. 107637, doi: 10.1016/j.compag.2023.107637.

[23] Y. Jiang, Z. Wei, and G. Hu, "An efficient detection method for tea leaf blight in UAV remote sensing images under intense lighting conditions based on MLDNet," *Comput. Electron. Agricult.*, vol. 229, Feb. 2025, Art. no. 109825, doi: 10.1016/j.compag.2024.109825.

[24] S. Elfwing, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural Netw.*, vol. 107, pp. 3–11, Jan. 2018.

[25] L. Lu, Q. Xiong, B. Xu, and D. Chu, "MixDehazeNet: Mix structure block for image dehazing network," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–10, doi: 10.1109/IJCNN60899.2024.10651326.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.

[26] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2021, *arXiv:2110.13389*.
[27] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," 2019, *arXiv:1911.08287*.
[28] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," 2017, *arXiv:1706.08500*.
[29] S. Barratt and R. Sharma, "A note on the inception score," 2018, *arXiv:1801.01973*.
[30] L. A. Gatys, A. S. Ecker, and M. Bethge, "Image style transfer using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2414–2423, doi: 10.1109/CVPR.2016.265.
[31] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 2242–2251, doi: 10.1109/ICCV.2017.244.
[32] L. Wen, C. Gao, and C. Zou, "CAP-VSTNet: Content affinity preserved versatile style transfer," 2023, *arXiv:2303.17867*.
[33] S. Zheng, P. Gao, P. Zhou, and J. Qin, "Puff-Net: Efficient style transfer with pure content and style feature fusion network," 2024, *arXiv:2405.19775*.
[34] T. Qi et al., "DEADiff: An efficient stylization diffusion model with disentangled representations," 2024, *arXiv:2403.06951*.
[35] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
[36] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.
[37] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.
[38] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017, doi: 10.1109/TPAMI.2016.2577031.
[39] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[40] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, no. 5605425, pp. 1–15, 2023.
[41] H. Zhang, K. Liu, Z. Gan, and G.-N. Zhu, "UAV-DETR: Efficient end-to-end object detection for unmanned aerial vehicle imagery," 2025, *arXiv:2501.01855*.
[42] Z. Zheng et al., "Enhancing geometric factors in model learning and inference for object detection and instance segmentation," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 8574–8586, Aug. 2022, doi: 10.1109/TCYB.2021.3095305.
[43] Z. Gevorgyan, "SIoU loss: More powerful learning for bounding box regression," 2022, *arXiv:2205.12740*.
[44] S. Ma and Y. Xu, "MPDIoU: A loss for efficient and accurate bounding box regression," 2023, *arXiv:2307.07662*.



Gensheng Hu (Member, IEEE) received the Ph.D. degree from South China University of Technology, Guangzhou, China, in 2006.

He is currently a Professor with the School of Electronics and Information Engineering, Anhui University, Hefei, China. His research interests include machine learning, image processing, and pattern recognition.



Zijing Wei received the B.S. degree in electrical engineering and automation from Anhui University, Hefei, China, in 2023, where he is currently pursuing the Ph.D. degree in electronic science and technology.

His research interests are computer vision and remote sensing image processing.



Yongcheng Jiang received the M.D. degree from Shanghai University, Shanghai, China, in 2005.

He is currently an Associate Professor with the School of Electronics and Information Engineering, Anhui University, Hefei, China. His research interests include control theory, image processing, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:42:27 UTC from IEEE Xplore. Restrictions apply.